CSA06507-Generative AI

Lab Experiment's

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Unit 2: Lab Programs for Prompting

Experiment 9: Python Code Generation Using Structured Prompting

Aim:

To design structured prompts that instruct a Large Language Model (LLM) to generate Python programs for specified computational problems and validate the generated code by executing it in Google Colab.

Algorithm:

1. Start the program.
2. Identify the given computational problem.
3. Design a structured prompt with the problem description, input, output, and constraints.
4. Instruct the LLM to generate Python code.
5. Copy the generated Python code into Google Colab.
6. Execute the code.
7. Check the output with the expected result.
8. Validate the code for errors and correctness.
9. Display the final output.
10. Stop the program.

Python Program

The following program calculates the factorial of a given number and validates the result using test cases.

Python

Run

Python Code Generation Using Structured Prompting

Step 1: Define the problem

Find the factorial of a given number.

Step 2: Structured Prompt

prompt = """

Write a Python program to calculate factorial of a number.

Requirements:

1. Accept an integer as input.

2. Calculate factorial using a loop.

3. Display the factorial.

4. Handle negative numbers.

5. Use simple Python code.

"""

```
print("STRUCTURED PROMPT:")
print(prompt)

# Step 3: Generate Python program

def factorial(n):
    if n < 0:
        return "Factorial is not defined"

    fact = 1
    for i in range(1, n + 1):
        fact = fact * i

    return fact

# Step 4: Get input
num = int(input("Enter a number: "))

# Step 5: Execute generated code
```

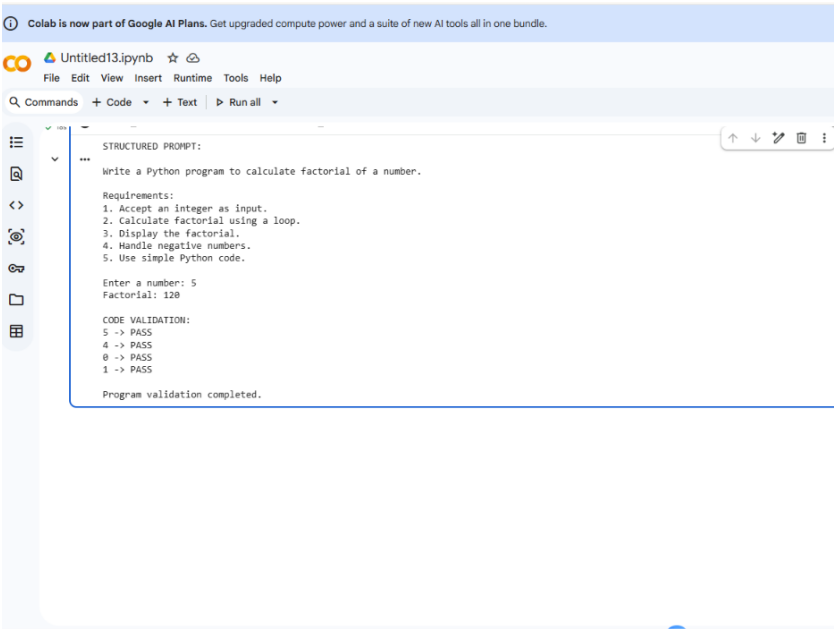
Input

Example input:

Plain text

Enter a number: 5

Output



Requirements:

1. Accept an integer as input.

2. Calculate factorial using a loop.

3. Display the factorial.

4. Handle negative numbers.

5. Use simple Python code.

Factorial: 120

CODE VALIDATION:

5 -> PASS

4 -> PASS

0 -> PASS

1 -> PASS

Program validation completed.

Result

Thus, Python code generation using structured prompting was successfully implemented in Google Colab. A structured prompt was designed to generate a factorial program, and the generated code was tested using multiple test cases. The program produced the expected outputs and passed the validation tests.